

MATH 100:

Linear Algebra

Yuqing Yang
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School of Computer Science and Engineering

09/02/2026

About me

Yuqing Yang, PhD

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- ▶ 2016 - 2020 B.Eng @ Nanjing University, China
- ▶ 2021 - 2025 PhD @ The Ohio State University, USA
- ▶ 2025 - 2026 PostDoc @ CISPA Helmholtz Center for Information Security, Germany
- ▶ 2026 - Now Assistant Professor @ Macau University of Science and Technology, Macau

Research and Services

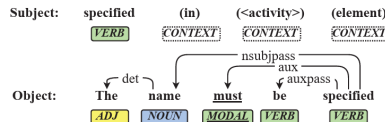
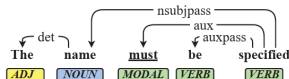
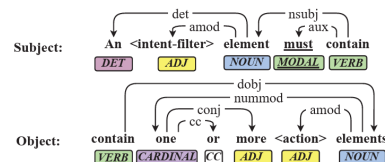
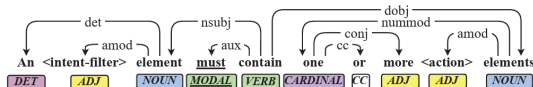
Cybersecurity

- ▶ Mobile Applications
- ▶ Web Applications
- ▶ Cloud Ecosystems
- ▶ Miniapps

2016-2020: Nanjing University, Software Engineering

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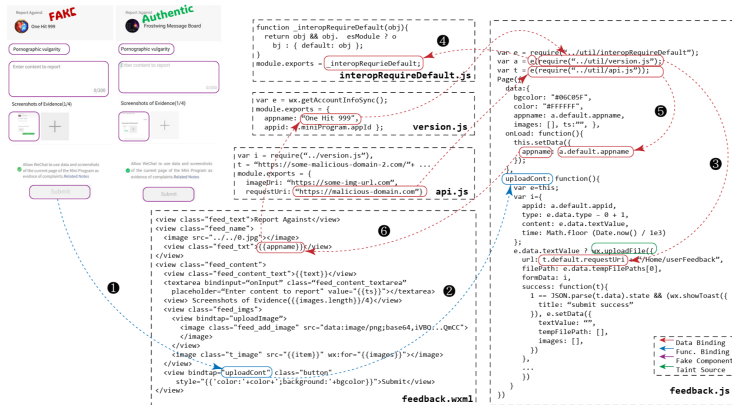
Software Requirement Retrieval



2021-2025: The Ohio State University, Computer Science and Engineering

2021-2025: The Ohio State University, Computer Science and Engineering

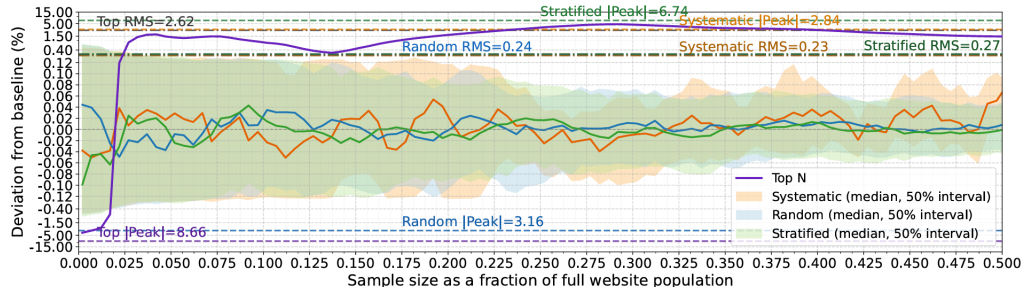
Miniapp Vulnerability and Malware



2025-2026: CISPA Helmholtz Center for Information Security

2025-2026: CISPA Helmholtz Center for Information Security

Web Security Measurement



2026-: M.U.S.T.

Agentic Ecosystem Security

1. 程序语言的语义映射差异

Java (静态编译型语言)

```
1 public class User{
2   public String getName() {...}
3 };
4 user.getName();
```

预先编译
静态访问

在Java中, 类型是明确定义的, 从字节码中可以判断其数据类型

JavaScript (动态解释型语言)

```
1 const user = {
2   getName: function() {...}
3 };
4 user.getName()
5 user["get"+"Name"]()
```

动态类型
运行时处理

在JavaScript中, 变量类型是动态决定的, 只有在运行时才能确定变量类型

启示: 编程语言的语义类型推断由语言特性决定, 需要针对性设计分析方法, 避免粗放解读导致错误理解。

2. 自然语言的语义组合方式差异

汉语: 分析语 (Analytic Language)

我

在

读

这个

文件

分析语特征: 语法功能主要依赖语序与上下文, 相对独立

英语: 曲折语/曲折倾向 (Fusional Language)

I

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read

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the

file

屈折语特征: 语法功能大量依赖词形变化, 词根附带语义

日语: 黏着语 (Agglutinative Language)

私

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黏着语特征: 语法功能通过后缀助词词尾, 逐层附加语义

启示: 自然语言的语义组合方式亦由语言特性决定, 针对大语言模型安全分析需要针对性设计语义分析与叠加规则。

Office Hour

My office is, surprisingly...

Office Hour

My office is, surprisingly...



Office Hour

My office is, surprisingly...



A5 Block

Office Hour

My office is, surprisingly...



A5 Block

Office hours:

► Tuesday: 0900 - 1100

Office Hour

My office is, surprisingly...



A5 Block

Office hours:

- ▶ Tuesday: 0900 - 1100
- ▶ Wednesday: 1430 - 1730

Office Hour

My office is, surprisingly. . .



A5 Block

Office hours:

- ▶ Tuesday: 0900 - 1100
- ▶ Wednesday: 1430 - 1730
- ▶ Thursday: 0900 - 1100

Office Hour

My office is, surprisingly...



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Office hours:

- ▶ Tuesday: 0900 - 1100
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- ▶ Friday: 1400 - 1700

Make an appointment!

Policies

- ▶ Language: English
- ▶ Communication:
 - ▶ Speak out!
 - ▶ or Stare at me...
 - ▶ or send me email or write in envelope at back of classroom

Schedule

Wednesday, 0900 - 1150

- ▶ In business travel: Oct. 26th & Nov. 18th
- ▶ Will schedule with you about that

Format

- ▶ Assignments
- ▶ Mid-term exam
- ▶ Final exam

Usage of AI

In Security: **USE, BUT VERIFY**

Usage of AI

In Security: **USE, BUT VERIFY**

- ▶ Verify with yourself

Usage of AI

In Security: **USE, BUT VERIFY**

- ▶ Verify with yourself
- ▶ Verify with classmates

Usage of AI

In Security: **USE, BUT VERIFY**

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Usage of AI

In Security: **USE, BUT VERIFY**

- ▶ Verify with yourself
- ▶ Verify with classmates
- ▶ Verify with me

Midterm and Final exam is closed-book, so be prepared.

How to Write an Email

How to Write an Email

- ▶ Title: [MATH100] What for Who

How to Write an Email

- ▶ Title: [MATH100] What for Who
- ▶ Body:

How to Write an Email

- ▶ Title: [MATH100] What for Who
- ▶ Body:
 - ▶ Introduce yourself!

How to Write an Email

- ▶ Title: [MATH100] What for Who
- ▶ Body:
 - ▶ Introduce yourself!
 - ▶ (optional) Itemize the needs

How to Write an Email

- ▶ Title: [MATH100] What for Who
- ▶ Body:
 - ▶ Introduce yourself!
 - ▶ (optional) Itemize the needs
 - ▶ Just casual is fine

How to Write an Email

- ▶ Title: [MATH100] What for Who
- ▶ Body:
 - ▶ Introduce yourself!
 - ▶ (optional) Itemize the needs
 - ▶ Just casual is fine
 - ▶ Check spelling

Boîte à Musique

Language

Language

Language

Language

Language

Language

Language

Language

Language

Language

Language